



AUGUST/RESIT EXAMINATIONS 2024/2025

MODULE: CSC1021 - Operating Systems

PROGRAMME(S):

CASE	BSc Computer Apps (Sft.Eng.)
COMSCI	BSc Computer Science
ECSA	Study Abroad Program EC
ECSAO	Study Abroad Program EC

STAGE: 1,2

EXAMINER(S):

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TIME ALLOWED: 2 Hours

INSTRUCTIONS: Answer all questions.

PLEASE DO NOT TURN OVER THIS PAGE UNTIL YOU ARE INSTRUCTED TO DO SO.
The use of programmable or text storing calculators is expressly forbidden.

There are no additional requirements for this paper.

QUESTION 1

[TOTAL MARKS: 25]

Q 1(a)

[5 Marks]

In the Linux process life-cycle, explain why a process may transition from a *ready* state to a *running* state and from a *running* state to a *ready* state.

Q 1(b)

[5 Marks]

A C program on Linux contains the following two lines of code:

```
execlp("/bin/ls","ls",NULL);  
printf("123");
```

When this program is run, will "123" be printed to the screen? Explain your reasoning, considering what happens when *execlp* is called.

Q 1(c)

[9 Marks]

Explain the memory layout of a C program using example C code. For each of the following memory sections: text, initialised data, uninitialised data, heap, and stack, provide one line of C code that places a variable or function in that section.

Q 1(d)

[6 Marks]

Explain what is meant by the terms *parent* and *child* when talking about processes in Linux. In your answer, describe any relevant system calls.

[End of Question 1]

QUESTION 2

[TOTAL MARKS: 25]

Q 2(a)

[7 Marks]

Using Amdahl's law, calculate the speedup of a program with a 50% serial proportion for 10 processors. Assuming there is an unlimited number of processors available, what's the maximum speedup that could be achieved?

Q 2(b)

[7 Marks]

Explain what busy waiting (spinlock) is and why it is undesirable. Then, using an example, describe a synchronization primitive a programmer could use to avoid it.

Q 2(c)

[6 Marks]

Message passing and shared memory are two ways that processes on a system can communicate. Explain what both of these are.

Q 2(d)

[5 Marks]

A thread scheduled using shortest-remaining-time-first (SRTF) will always run to completion without being pre-empted. Is this statement true or false? Justify your answer.

[End of Question 2]

QUESTION 3

[TOTAL MARKS: 25]

Q 3(a)

[8 Marks]

A programmer is attempting to implement a solution to the bounded-buffer/producer-consumer problem for a buffer of size **n**. They have encountered difficulty when using the implementation below. Three semaphores **mutex**, **full** and, **empty** have been initialised to **1, 0, and n**, respectively.

The following implementation has logical issues. Outline the changes you would make for it to operate correctly. Refer to specific line numbers in your answer, and explain the correct use of each semaphore.

Code for Q3(a)	
Consumer process:	Producer Process:
<pre> 01: while (true) { 02: wait(empty); 03: signal(mutex); 04: ... 05: /* remove item from buffer*/ 06: ... 07: signal(full); 08: signal(empty); 09: }</pre>	<pre> 10: while (true) { 11: /* produce an item */ 12: signal(empty); 13: wait(mutex); 14: ... 15: /* add produced item to buffer */ 16: ... 17: wait(empty); 18: }</pre>

Q 3(b)

[11 Marks]

The program below has two threads that may execute concurrently (i.e. *thread-A* and *thread-B*), and two locks shared between these threads (i.e. *lock1* and *lock2*). Using the program below, explain the concept of deadlock. In your answer, describe the four necessary conditions for deadlock and show how they apply with reference to specific line numbers.

Code for Q3(b)	
code from thread-A	code from thread-B
<pre> A-1: lock1.lock() A-2: lock2.lock() A-3: a=a+b A-4: lock1.unlock() A-5: lock2.unlock()</pre>	<pre> B-1: lock2.lock() B-2: lock1.lock() B-3: b=b+a B-4: lock2.unlock() B-5: lock1.unlock()</pre>

Q 3(c)**[6 Marks]**

Describe two different changes that could be made to the program for Q3(b) so that deadlock would not occur. In your answer, indicate what condition(s) of deadlock each change invalidates.

[End of Question 3]

QUESTION 4**[TOTAL MARKS: 25]****Q 4(a)****[4 Marks]**

The following sequence of commands are run on Linux:

```
echo "123" > myfile1.txt
ln myfile1.txt myfile3.txt
ln myfile1.txt myfile2.txt
echo "XYZ" > myfile2.txt
rm myfile2.txt
echo "ABC" > myfile2.txt
```

After the above commands are executed, what content is stored in myfile1.txt?

Q 4(b)**[6 Marks]**

Explain what is meant by the term DMA (Direct Memory Access). In your answer, explain why DMA would be used instead of programmed I/O.

Q 4(c)**[12 Marks]**

Paging is a memory management scheme.

- i) Explain how a *page table* works, using an example of a 32-bit virtual address.
- ii) Describe the role of the Translation Lookaside Buffer (TLB) in paging.

Q 4(d)**[3 Marks]**

Explain what the following Linux command does:

```
chmod 755 myfile
```

[End of Question 4]

[END OF EXAM]